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Strategic Defect Engineering Enabled Efficient Oxygen Evolution Reaction in Reconstructed Metal-Organic Frameworks

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Metal-organic frameworks (MOFs) have emerged as promising pre-catalysts for oxygen evolution reaction (OER) due to their marvelous structural reconstruction process in strongly alkaline media. However, targeting design MOF structures to achieve excellent OER performance of reconstructed products is a challenge. Here, a strategic defect engineering is used to promote the OER performance of reconstructed products. Briefly, modified linkers with monocarboxylic acids (ferrocene carboxylic acid, FcCA) are incorporated into MOF (NiBDC-FcCA), leading to its stepwise reconstruction into Fe-doped $Ni(OH)_2$ and NiOOH during the OER process, with the oxygen vacancy and strategic doping of metal Fe persisting throughout the multi-step reconstruction. Benefiting from the synergistic interaction of oxygen vacancies and Fe doping, NiBDC-FcCA delivers the extremely enhanced current density at 1.6 V versus reversible hydrogen electrode by ≈9 times compared with that of NiBDC. Moreover, the optimized NiBDC-FcCA/Fe foam exhibits excellent OER catalytic activity and stability with a low overpotential of 250 mV at 200 mA cm^{-2} and negligible activity decay after 1200 h at 1 A cm^{-2}. Density function theory calculations reveal that Fe doping weakens the interaction of oxygen intermediate with Ni sites, favoring the formation of OOH* to accelerate the OER process.

1. Introduction

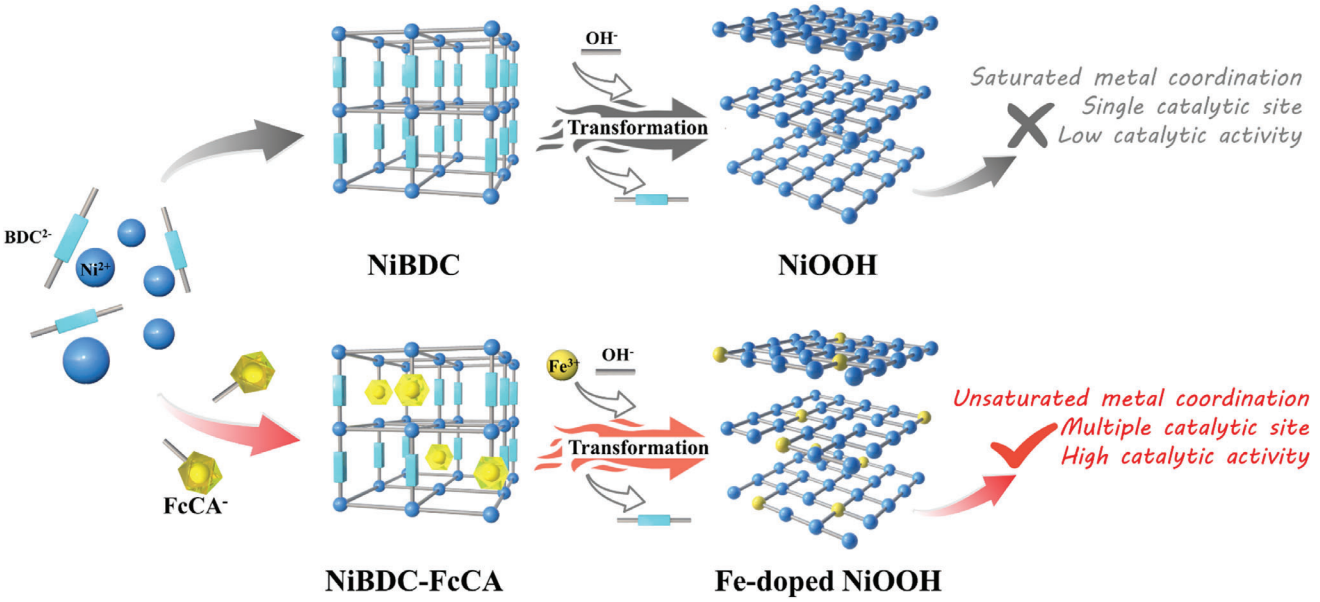
Metal-organic frameworks (MOFs), a type of promising porous crystalline materials formed from metal nodes and organic ligands via coordination bonds, have become the research frontier in the realm of electrochemical energy storage and conversion taking advantage of long-range order, structural variability, and tailorable porosity.[1] Especially for oxygen evolution reaction (OER), deemed as the most challenging step in electrochemical water splitting due to the complex electron transfer process,[2] this category of materials has garnered emerging attention and become feasible platforms for developing efficient OER electrocatalysts. In this regard, the advanced design feature at the atomic/molecular level can render these materials with improved OER activity through rational design of synthetic protocol.[3] Sequentially, a series of state-of-the-art MOF-based OER electrocatalysts with satisfactory OER performance have been achieved, represented by zeolitic imidazolate-based MOFs and carboxylic acid-based MOFs, which are prized for their facile synthesis, permanent porosity, abundant active sites, high design flexibility.[4]

Research has shown that introducing defect structures to MOFs could significantly enhance the OER activity of targeted catalysts.[3b,5] This approach helps alter the coordination environment of metal centers, expose more active sites, and promote charge transfer, thereby improving the OER performance. For example, introducing missing linkers such as ferrocene carboxylic acid (FcCA) to form defective MOFs (CoBDC Fc, Fc-NiCo-BDC, BDC = 1,4-dicarboxybenzene) can modulate the electronic structures of the active center, which further optimizes the adsorption/desorption strength of reaction intermediates at the active center, thereby enhancing the OER activity.[3b,5a] Additionally, partial substitution of FcCA in the CoBDC enables the active center with distinctive functional sites, including Co1 sites coordinated to the oxygen atoms from carboxyl groups of FcCA, unsaturated coordinated Co2 sites generated by the missing ligand, and Fe sites of FcCA. These active sites associated with ultrathin structure and tensile strain allow CoBDC FcCA catalysts with improved OER performance.[5b] Previous studies have also shown that Fc-based MOF pre-electrocatalysts constructed from ferrocene dicarboxylic acid (FcDA) ligands have also been proven to be a promising class of efficient OER catalysts.[6] For example, a series of FcDA-based MOFs (CoNiFc-MOFs) are produced via

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Scheme 1. Schematic illustration of the introduction of the defect in defect-free MOF through FcCA modification for promoting the OER.

the facile solvothermal method.[6b] These catalysts exhibit superior OER activity, and tuning the Ni amount can further improve the OER activity. These significant works have aroused considerable interest in obtaining MOF-based catalysts with excellent OER performance.

Besides, it was demonstrated a concept that the MOFs with moderate structural bond strengths, especially for carboxylic acid-based MOF, are amenable to the occurrence of partial or complete structural transformations, operating in strongly alkaline media during OER, forming metal hydroxides/hydroxyl oxides that serve as the true active species.[7] The in situ reconstruction process typically involves several variations, such as the increase of active sites, doping of residual cations, the appearance of unsaturated coordination sites, and the formation of heterogeneous interfaces, etc, which substantially affect the nature of reconstructed products.[8] Therefore, compared to directly synthesized metal hydroxides/hydroxyl oxides, the reconstructed species generally exhibit superior OER activity despite their similar chemical properties.[9] In this situation, the in situ structural reconstruction between MOFs and metal hydroxides provides significant opportunities to identify the possible constituents and architectural features for preparing reconstructed products from the most fundamental starting point. However, most of studies are focused on identifying the reconstruction phenomena and the real-time tracking of fine-scale changes in the catalyst structure during the reaction process is lacking. The impact of defects on catalyst reconstruction and intrinsic activity is still not fully understood.

Herein, we proposed a defect engineering for boosting the OER, in that, breaking the regular periodic arrangement of the organic linker in periodic bulk structure, which accesses the anticipated reconfiguration products capable of inheriting the nature of the pre-catalyst MOF (**Scheme 1**). Taking the NiBDC consisting of the Ni–OH layer and BDC^{2-} pillar as an example, order manipulation of the inner framework, introducing modified linker defects with monocarboxylic acids (FcCA) into defect-free NiBDC, offers particularly fascinating abundant and highly-accessible unsaturated sites because of ligand vacancies. Additionally, strategic doping of metal Fe efficiently modulates the electronic structure. Benefiting from the synergistic interaction of unsaturated sites and Fe doping, NiBDC-FcCA delivers the enhanced current density at 1.6 V versus reversible hydrogen electrode (vs. RHE) by ≈9 times compared with that of NiBDC. The structural changes of this material were explored using in situ Raman and electron paramagnetic resonance (EPR) spectra measurements. The optimized catalyst displays outstanding OER performance, requiring only 250 mV overpotential at 200 mA cm^{-2} and maintaining activity for up to 1200 h at a continuous current of 1 A cm^{-2}. Anion exchange membrane water electrolysis (AEMWE) cells consisting NiBDC-FcCA/Fe foam as an anode catalyst deliver an excellent OER activity, requiring only a low cell voltage of 1.85 V to achieve a current density of 1 A cm^{-2}, surpassing the commercial Pt/C||RuO_2 catalyst (1.92 V). Density function theory (DFT) calculations unravel that Fe doping weakens the adsorption of the oxygen intermediate with Ni sites, favoring the formation of OOH*, and thus resulting in higher catalytic activity. This work, by demonstrating the feasibility of a strategic defect engineering, suggests that harnessing the designed MOFs as pre-catalysts is a promising strategy for the directional synthesis of novel and efficient OER catalysts.

2. Results and Discussion

As a proof-of-concept study, nickel-based MOF (NiBDC) that serves as an instance was first prepared by solvothermal reaction method using BDC ligand. As an optimization, defective MOF (NiBDC-FcCA) was prepared by partially substituting BDC with FcCA with modified linkers due to missing the monocarboxylic acid group, aiming to simultaneously create the Fe doping and structural defects for the resulting

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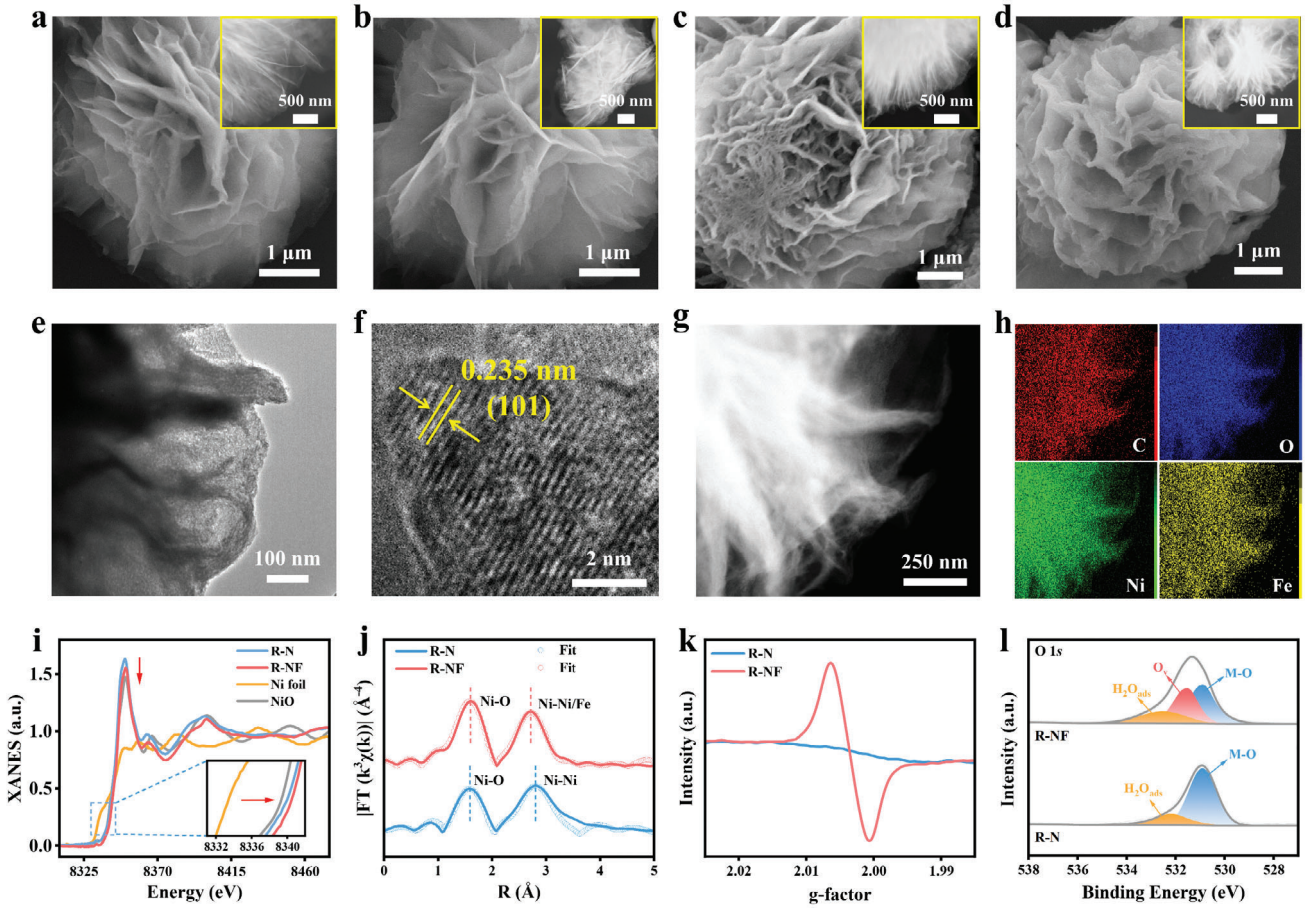


Figure 1. The morphology characterization and structural analysis of samples. SEM images of a) NiBDC, b) NiBDC-FcCA, c) R-N, and d) R-NF. The insets show the corresponding scanning transmission electron microscopy (STEM) images. e) TEM image of R-NF. f) HR-TEM of R-NF. g), h) STEM and corresponding EDX mapping of R-NF. i) XANES spectra at Ni *K*-edge of R-NF and R-N, Ni foil, and NiO. j), k), l) Fourier transformed EXAFS spectra, EPR spectra, and O 1*s* spectra of R-NF and R-N.

reconstruction product. The powder X-ray diffraction (PXRD) patterns reveal a high consistency in the diffraction peaks between NiBDC-FcCA and NiBDC (Figure S1, Supporting Information), indicating that the introduction of FcCA without damaging the overall structural integrity of NiBDC. The scanning electron microscopy (SEM) and transmission electron microscopy (TEM) images give the smooth surface nanoflower structures of both NiBDC and NiBDC-FcCA (**Figure 1**a,b; Figure S2, Supporting Information). Furthermore, energy-dispersive X-ray spectroscopy (EDX) mapping images of NiBDC-FcCA show the even distribution of elements in the nanoflowers (Figures S3 and S4, Supporting Information), confirming the homogeneous doping of FcCA in the NiBDC. It has been demonstrated that the instability of carboxylic acid-based MOF materials in alkaline media,[9d,10] a detailed investigation was conducted on the reconstruction of the sample. Initially, NiBDC and NiBDC-FcCA were immersed into 1.0 M KOH electrolyte separately, and the BDC^{2-} concentration of the electrolyte was detected by UV-absorption spectroscopy at various time intervals to evaluate the degradation rate of the samples in the electrolyte. As shown in Figure S5 (Supporting Information), both NiBDC and NiBDC-FcCA powders were destroyed within ≈5 min. Therefore, NiBDC and NiBDC-FcCA were immersed in 1.0 M KOH for sufficiently 1.0 h to achieve a complete reconstruction of the materials at static KOH solution. The completed reconstruction products of NiBDC and NiBDC-FCCA are named R-N and R-NF, respectively. Then, a series of characterizations were performed to detect the reconstructed samples. The PXRD patterns of R-N and R-NF (Figure S6, Supporting Information) indicate the disappearance of diffraction peaks assigned to MOFs, with all peaks aligning well with $Ni(OH)_2$ (JCPDS *No.* 14–0117). The high-resolution TEM (HR-TEM) of R-NF displays a lattice distance of 0.235 nm, corresponding to the (101) crystal plane of $Ni(OH)_2$ (Figure 1f).[10a] These observations confirm the complete reconstruction of NiBDC-FcCA into metal hydroxide in an alkaline electrolyte. According to the SEM and TEM images of R-N and R-NF, the morphologies of reconstructed samples remain essentially unchanged but exhibit significantly rougher surfaces compared to NiBDC and NiBDC-FcCA (Figure 1c–e; Figure S7, Supporting Information). Meanwhile, changes in FcCA are investigated by the comparison of proton nuclear magnetic resonance (^{1}H NMR) spectra before and after sample reconstruction (Figure S9, Supporting Information). The absence of the FcCA signal in the ^{1}H NMR spectrum of R-NF indicates the absence of FcCA residue in the sample after reconstitution.

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Additionally, EDX mapping images exhibit the even distribution of C, O, Ni, and Fe elements (Figure 1g,h; Figure S8, Supporting Information). The metal contents in R-NF (Ni: 64.9 wt%; Fe: 1.93 wt%) are quantified using inductively coupled plasma optical emission spectroscopy (ICP-OES) (Table S1, Supporting Information). These results confirm that NiBDC-FcCA reconstructs to Fe-doped $Ni(OH)_2$ in an alkaline electrolyte.

To investigate the effect of the deliberate introduction of FcCA on the electronic states of pre-catalyst and derivative reconstruction products, we carried out the EPR spectroscopy, X-ray photoelectron spectroscopy (XPS) and X-ray absorption spectroscopy (XAS) tests. In the EPR spectra, NiBDC-FcCA shows a characteristic signal at g = 2.003, implying the presence of abundant oxygen vacancies within defective NiBDC-FcCA derived from the introduction of FcCA (Figure S10, Supporting Information).[11] Similarly, the consistent signal output from the reconstructed product R-NF suggests the inheritance of abundant oxygen vacancies (Figure 1k). The Ni 2*p* XPS spectra of NiBDC and NiBDC-FcCA (Figure S11a, Supporting Information) show two peaks at ≈856.0 and 873.9 eV assigned to Ni $2p_{3/2}$ and Ni $2p_{1/2}$, respectively, suggesting the presence of Ni^{2+} species.[12] Notably, the introduction of FcCA results in a higher binding energy of Ni 2*p* in NiBDC-FcCA compared to NiBDC, which is attributed to the creation of unsaturated Ni metal sites with a more positive valence in NiBDC-FcCA. Following structural reconstruction, a pair of new characteristic peaks near 856.8 and 874.6 eV emerged in Ni 2*p* of R-NF and R-N (Figure S11c, Supporting Information), which is attributed to Ni^{3+} species.[10b]

The local structural information of samples was analyzed using XAS. The Ni K-edge X-ray absorption near-edge structure (XANES) spectra display that the energy positions of the pre-edge peaks of R-N and R-NF are higher than those of Ni foil and NiO, indicating that the valence state of Ni is higher than 2+, consistent with the results of XPS. And the energy position of the pre-edge peak of R-NF is shifted to higher energies than R-N (Figure 1i). This result verifies the presence of a higher valent state of Ni within R-NF.[9c] By contrast, the white line peak of R-NF is slightly reduced compared to R-N, indicating that the local coordination environments of Ni in R-NF and R-N are distinct. For an in-depth study of variations in the coordination environment of nickel atoms, the Fourier transform of the extended X-ray absorption fine structure (EXAFS) spectra of R-NF and R-N have been curve-fitted (Figure S12, Supporting Information). As shown in Figure 1j, the bond distance of Ni—O in R-NF is 2.06 Å, nearly identical to R-N. On the other hand, the Ni—O coordination number of R-NF is 5.9, which is smaller than that of R-N (6.2), indicating a decrease in the number of coordinated oxygens in R-NF, giving rise to more unsaturated metal sites. Meanwhile, compared to NiBDC and R-N, the O 1*s* XPS spectra of NiBDC-FcCA and R-NF (Figures 1l; Figure S11b, Supporting Information) show a new characteristic peak attributed to oxygen vacancies (O_V), consistent with the results of fitted EXAFS.[13] Notably, a significant presence of Fe^{3+} in R-NF, according to the Fe 2*p* XPS spectra of R-NF, is observed (Figure S11d, Supporting Information), indicating the release of Fe^{3+} from FcCA in 1.0 M KOH electrolyte,[6a,9c] and then the free Fe^{3+} actively engages in the in situ structural reconstruction to produce Fe-doped $Ni(OH)_2$. These results suggest that the strategic defect engineering achieves the effect of "killing three birds with one stone": 1) Oxygen vacancies generated by introducing FcCA in pristine MOF were inherited across reconfiguration; 2) Fe^{3+} is released by FcCA during the reconstruction process, which achieves its in situ doping in $Ni(OH)_2$ and generates new Fe catalytic sites around neighboring Ni sites; 3) These features in terms of oxygen vacancies and Fe doping contribute to the redistribution of the electronic structure, which is beneficial for improving the OER efficiency.

To more precisely investigate the intrinsic activity of prepared catalysts, the electrocatalytic OER performance of the prepared NiBDC-FcCA and reference samples including NiBDC, NiBDC-BA (where benzoic acid replaced FcCA), and RuO_2 were systematically evaluated under identical condition. The LSV curves show that NiBDC-FcCA exhibits superior OER activity, delivering the overpotentials of 332 mV at current densities of 100 mA cm^{-2}, respectively (**Figure 2**a,c), which is significantly lower than that of NiBDC (458 mV), NiBDC-BA (413 mV), and comparable to commercial RuO_2 (355 mV). Moreover, NiBDC-FcCA delivers the enhanced current density at 1.6 V vs. RHE by ≈9, 5, and 2 times compared with that of NiBDC, NiBDC-BA, and RuO_2, respectively. Such results indicate that the introduction of modified linker FcCA significantly enhances the catalytic activity of NiBDC. Additionally, Tafel slopes were derived from LSV curves to gain insights into the electrochemical reaction kinetics of the samples. As shown in Figure 2b, NiBDC-FcCA (46.3 mV dec^{-1}) has a smaller Tafel slope than NiBDC-BA (79.6 mV dec^{-1}), NiBDC (92.6 mV dec^{-1}), and commercial RuO_2 (76.4 mV dec^{-1}), indicating the maximum kinetic speed. Moreover, CV curves of NiBDC-FcCA and NiBDC were recorded in the potential range of 0.1 to 0.5 V (vs. Ag/AgCl) to investigate the electrochemical behavior of the samples (Figure S13, Supporting Information). The pair of redox peaks corresponds to the redox reaction of Ni^{2+}/Ni^{3+}. Compared to NiBDC, the redox peaks of NiBDC-FcCA undergo a slight positive shift, indicating that the incorporation of FcCA increases the potential at which Ni^{2+}/Ni^{3+} redox occurs.[14]

To understand the mechanism of NiBDC-FcCA with optimal OER activity, electrochemical impedance spectroscopy (EIS) was first carried out to further detect the charge-transfer process. NiBDC-FcCA shows a smaller semicircle than that of NiBDC and NiBDC-BA (Figure 2e), indicating rapid electronic transport of NiBDC-FcCA. The electrochemical active surface area (ECSA) of the catalyst can be evaluated by its double-layer capacitance (C_{dl}) value from the CV curves at different sweep rates in the non-Faraday interval. The NiBDC-FcCA possesses the highest C_{dl} value of 39.6 mF cm^{-2}, surpassing that of NiBDC (23.9 mF cm^{-2}) and NiBDC-BA (35.1 mF cm^{-2}), indicating that FcCA doping delivers more abundant active sites (Figures 2d; Figure S14, Supporting Information). In addition, the LSV curves were normalized by ECSA to evaluate the intrinsic activity of the samples. After ECSA normalization, the overpotential and Tafel slopes of the NiBDC-FcCA are still smaller than that of NiBDC, NiBDC-BA, and RuO_2 (Figures 2f; Figure S15, Supporting Information). Compared to benzoic acid, the use of FcCA as a dopant not only introduces oxygen vacancies but also incorporates doped Fe, which facilitates the redistribution of the electronic structure and significantly enhances OER performance. These illustrate the vital contribution of Fe doping to the improvement of OER activity.

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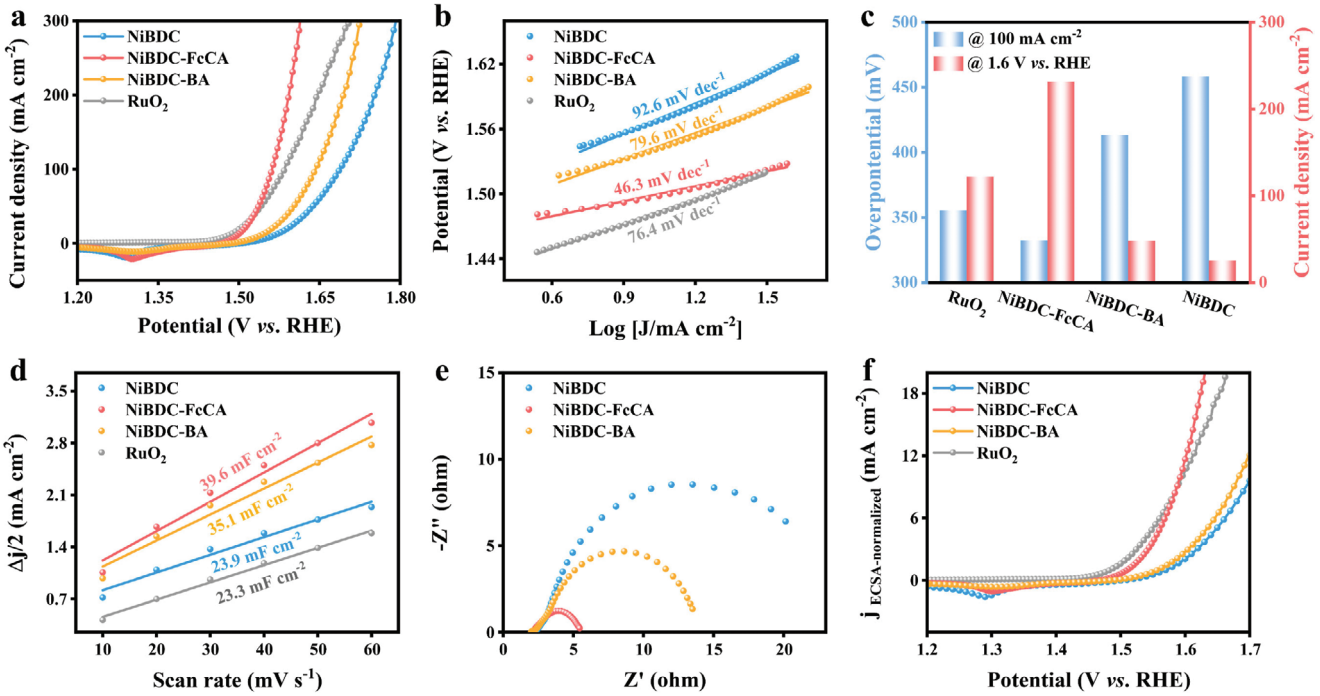


Figure 2. Electrocatalytic performance evaluation of samples. a) LSV curves. b) Tafel slopes. c) Comparison of overpotential at 100 mA cm^{-2} and current density at 1.6 V vs. RHE. d) Current densities at various scan rates. e) EIS. f) LSV curves normalized by ECSA.

Additionally, the impact of varying FcCA doping amounts on the electrocatalytic properties of the samples was investigated (Figures S16-18, Supporting Information). As shown in Figures S19 and S20 (Supporting Information), the lowest overpotential and Tafel slopes were observed at an FcCA/BDC ratio of 0.15. To further highlight the advantages of MOFs as an OER pre-catalyst, NiFe layered double hydroxides (NiFe LDH) with an identical Ni/Fe ratio as NiBDC-FcCA, and $Ni(OH)_2$-FcCA derived from direct FcCA doping with $Ni(OH)_2$, were synthesized. Compared with NiFe LDH and $Ni(OH)_2$-FcCA, NiBDC-FcCA still presents the optimal OER activity (Figures S21-S23, Supporting Information), further demonstrating the advantages of structural inheritance of reconstruction products for OER performance optimization.

In situ Raman testing was conducted to further investigate the reconstruction process of the MOFs pre-catalyst during OER. When the potential increases to 0.4 V, NiBDC exhibits two distinct Raman peaks near 480 and 556 cm^{-1}, corresponding to the E_g bending vibration mode and A_{1g} stretching vibration mode of Ni^{III}-O in NiOOH, respectively (**Figure 3b**). Compared with NiBDC, the Raman peak of Ni^{III}-O in NiBDC-FcCA appears at a higher potential (Figure 3c), indicating that the Ni species in NiBDC-FcCA are more difficult to be oxidized, consistent with the CV curves. Furthermore, the intensity ratio of two Ni^{III}-O Raman peaks (I_{556}/I_{480}) can assess the structural disorder of NiOOH.[15] As shown in Figure 3d, the I_{556}/I_{480} of NiBDC-FcCA is higher than that of NiBDC, indicating that the NiOOH catalyst reconstructed by NiBDC-FcCA presents numerous defects. To study the electronic state changes of MOFs after OER, XPS and EPR spectra of post-OER for NiBDC and NiBDC-FcCA were characterized. It was found that the Ni^{3+}/Ni^{2+} ratio of post-OER for NiBDC-FcCA and NiBDC increases compared to R-NF and R-N (Figure 3e,f), further indicating the formation of NiOOH during the electrocatalytic oxidation process. The EPR spectra show that post-OER NiBDC-FcCA possesses abundant oxygen vacancies (Figure 3g). Based on the above results and previous analysis, it is evident that NiBDC-FcCA underwent a multistep reconstruction, and importantly, oxygen vacancies were preserved in the multi-step reconstruction. Initially, NiBDC-FcCA underwent reconstruction into Fe-doped $Ni(OH)_2$ with plentiful oxygen vacancies in the alkaline electrolyte, followed by the in situ formation of Fe-doped NiOOH with plentiful oxygen vacancies during the OER process (Figure 3a).

To investigate the effect of Fe doping on enhancing the electrocatalytic activity of OER, DFT calculations on the geometrically optimized model structures of both NiOOH with oxygen vacancies (NiOOH-O_V) and Fe-doped NiOOH with oxygen vacancies (Fe-NiOOH-O_V) were carried out (Figure S24, Supporting Information). It is well known that the ΔG of the active sites and intermediates (OH*, O*, OOH*) is closely linked to the catalyst's activity.[6a,16] First, the geometrical structures of the intermediates adsorbed on the active Ni sites of NiOOH-O_V, and Ni, Fe sites of Fe-NiOOH-O_V were optimized and calculated (Figures S25-S27, Supporting Information). **Figure 4a,b** show the ΔG diagrams without ($U = 0$ V) and with ($U = 1.23$ V) the bias potential, respectively. It can be seen that the rate-determining step (RDS) of these three reaction pathways is the formation of OOH*. Remarkably, the ΔG of this step at the Ni sites decreased significantly after Fe doping, primarily due to the weakened adsorption of O* and OOH* at the Ni sites. Additionally, Fe sites in Fe-NiOOH-O_V exhibit higher catalytic

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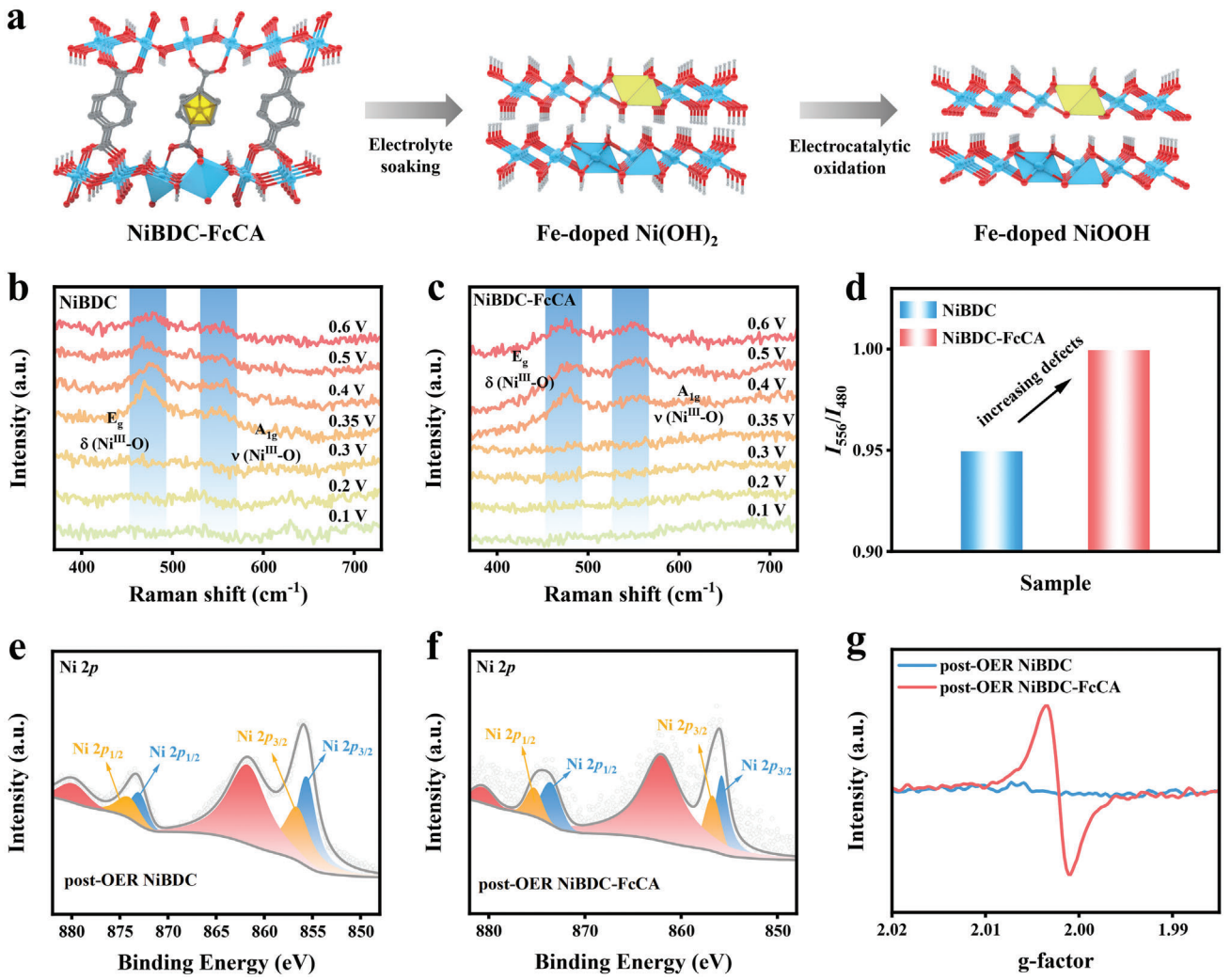


Figure 3. a) Schematic diagram of the structural reconstruction process of NiBDC-FcCA in OER progress. In situ Raman spectra of b) NiBDC and c) NiBDC-FcCA. d) The I_{556}/I_{480} of NiBDC and NiBDC-FcCA. Ni 2*p* spectra of post-OER for e) NiBDC and f) NiBDC-FcCA. g) EPR spectra of post-OER for NiBDC-FcCA and NiBDC.

activity than the Ni sites. Therefore, Fe doping not only enhances the catalytic activity of Ni but also introduces new Fe catalytic sites.

The electronic structures of NiOOH-O_V and Fe-NiOOH-O_V were investigated using partial density of states (PDOS). As shown in Figure 4c, g, the *d*-band center (ϵ_d) of Ni sites decreases from −1.48 eV for NiOOH-O_V to −1.98 eV for Fe-NiOOH-O_V, indicating the weakened interaction of oxygen intermediates with Ni sites on Fe-NiOOH-O_V. To further understand the effect of Fe doping, the crystal orbital Hamilton populations (COHP) were performed, with negative and positive -COHP values representing anti-bonding and bonding states, respectively.[17] The integral of -COHP (I) calculations indicates that doped Fe decreased the binding strength between the active site and OOH* intermediate (Figure 4d–f). The electron transfer of OOH*/catalyst interface was further analyzed through differential charge densities and Bader charge analysis. Compared to NiOOH-O_V (Ni), Fe-NiOOH-O_V (Ni) and Fe-NiOOH-O_V (Fe) exhibit weaker electron transfer intensity (Figure 4h–j), indicating their weaker adsorption for OOH*. Based on these results, it is demonstrated that Fe doping weakens the adsorption of oxygen intermediate with Ni sites and reduces the energy barrier of RDS, leading to superior catalytic activity.

The superior electrochemical activity of NiBDC-FcCA drove us to investigate its potential for industrial application feasibility. NiBDC-FcCA was grown in situ on Fe foam with excellent electrical conductivity and a large specific surface area, denoted as NiBDC-FcCA/Fe foam. Optical images depict the color change in Fe foam upon loading NiBDC-FcCA (Figure S28, Supporting Information). The PXRD pattern of NiBDC-FcCA/Fe foam closely matches that of NiBDC-FcCA, confirming successful synthesis and loading onto Fe foam (Figure S29, Supporting Information). The optimized electrocatalyst NiBDC-FcCA/Fe foam displays outstanding OER performance, requiring only 250 mV overpotential at 200 mA cm^{-2} and low Tafel slopes of 26.3 mV dec^{-1} (**Figures** 5a and Figure S30, Supporting Information). As

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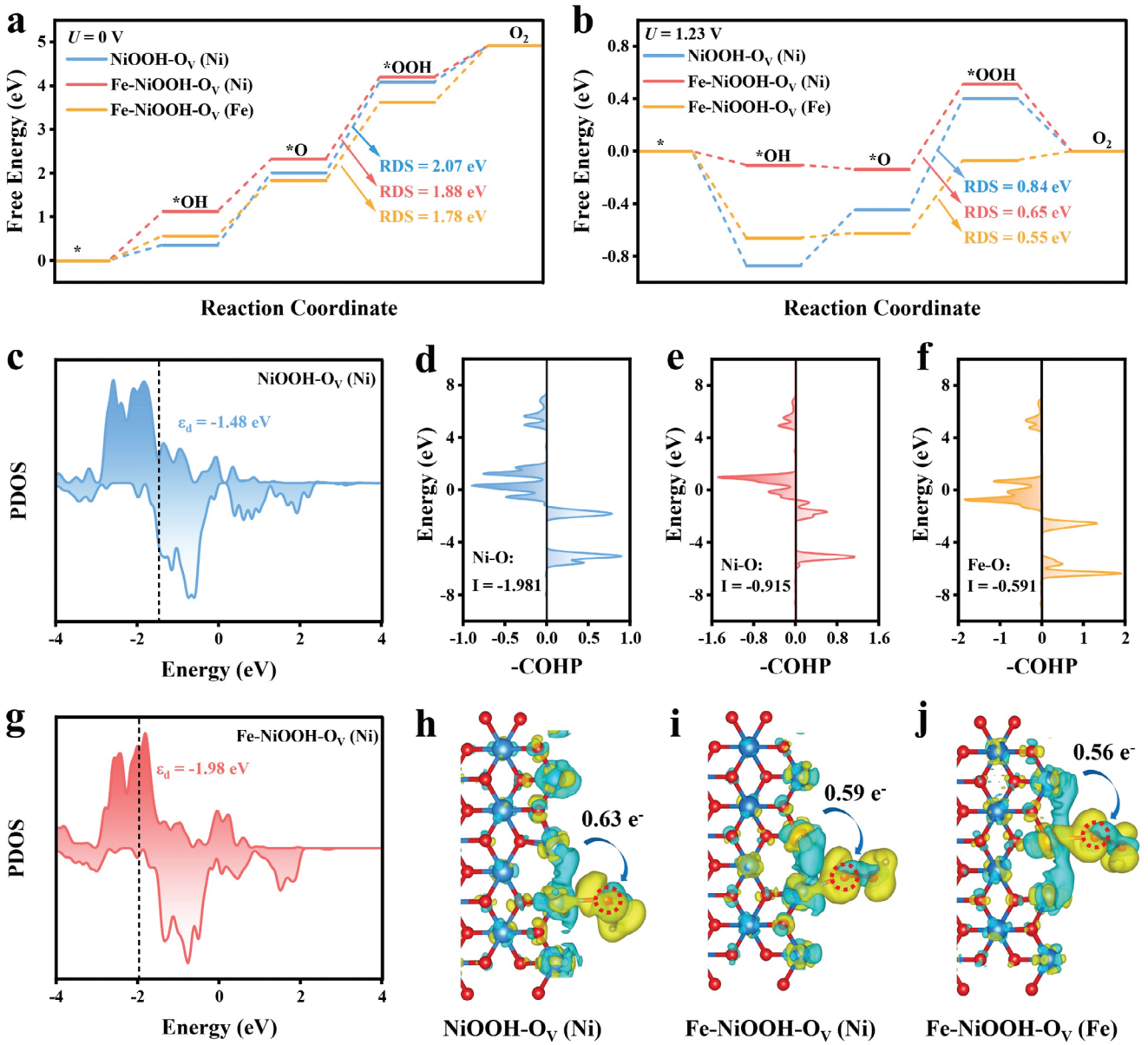


Figure 4. Calculated free energy diagrams of OER on NiOOH-O_V (Ni), Fe-NiOOH-O_V (Ni), and Fe-NiOOH-O_V (Fe) with a) $U = 0$ V and b) $U = 1.23$ V in 1 M KOH. PDOS of c) NiOOH-O_V (Ni) and g) Fe-NiOOH-O_V (Ni). COHP of M—O bond with the adsorption of *OOH intermediates for d) NiOOH-O_V (Ni), e) Fe-NiOOH-O_V (Ni), and f) Fe-NiOOH-O_V (Fe). h), i), j) The local structures and charge difference maps for OOH* adsorption. The yellow and blue isosurfaces represent electron gain and loss, respectively.

shown in Figure S31 (Supporting Information), the C_{dl} value of the NiBDC-FcCA/Fe foam (12.3 mF cm^{-2}) surpasses that of Fe foam (2.9 mF cm^{-2}), indicating that NiBDC-FcCA/Fe foam has more abundant active sites. The long-term stability of NiBDC-FcCA/Fe foam was also investigated by the chronopotentiometric *v-t* test. As shown in Figure 5d, the NiBDC-FcCA/Fe foam exhibits excellent stability toward OER, which can keep the current density to maintain a steady-state value for 1200 h at 1 A cm^{-2}. SEM images of NiBDC-FcCA/Fe foam show that the morphology of nanorod arrays has almost no change before and after *v-t* test (Figures S32 and S33, Supporting Information). The excellent catalytic performance of NiBDC-FcCA/Fe foam was further highlighted through comparisons with reported nickel-based catalysts (Table S3, Supporting Information).

Encouraged by the promising catalytic activity and stability exhibited by NiBDC-FcCA/Fe foam, an AEMWE cell was assembled using membrane electrode assemblies with NiBDC-FcCA/Fe foam as the anode and NiMo/Ni foam as the cathode under simulated industrial conditions (Figure 5c). The NiBDC-FcCA/Fe foam||NiMo/Ni foam requires only a low cell voltage of 1.85 V at a current density of 1 A cm^{-2} (Figure 5b), surpassing even the commercial catalyst Pt/C||RuO_2 (1.92 V) and currently reported AEMWE cells (Figure 5e; Table S4, Supporting Information). Moreover, as shown in Figure S34

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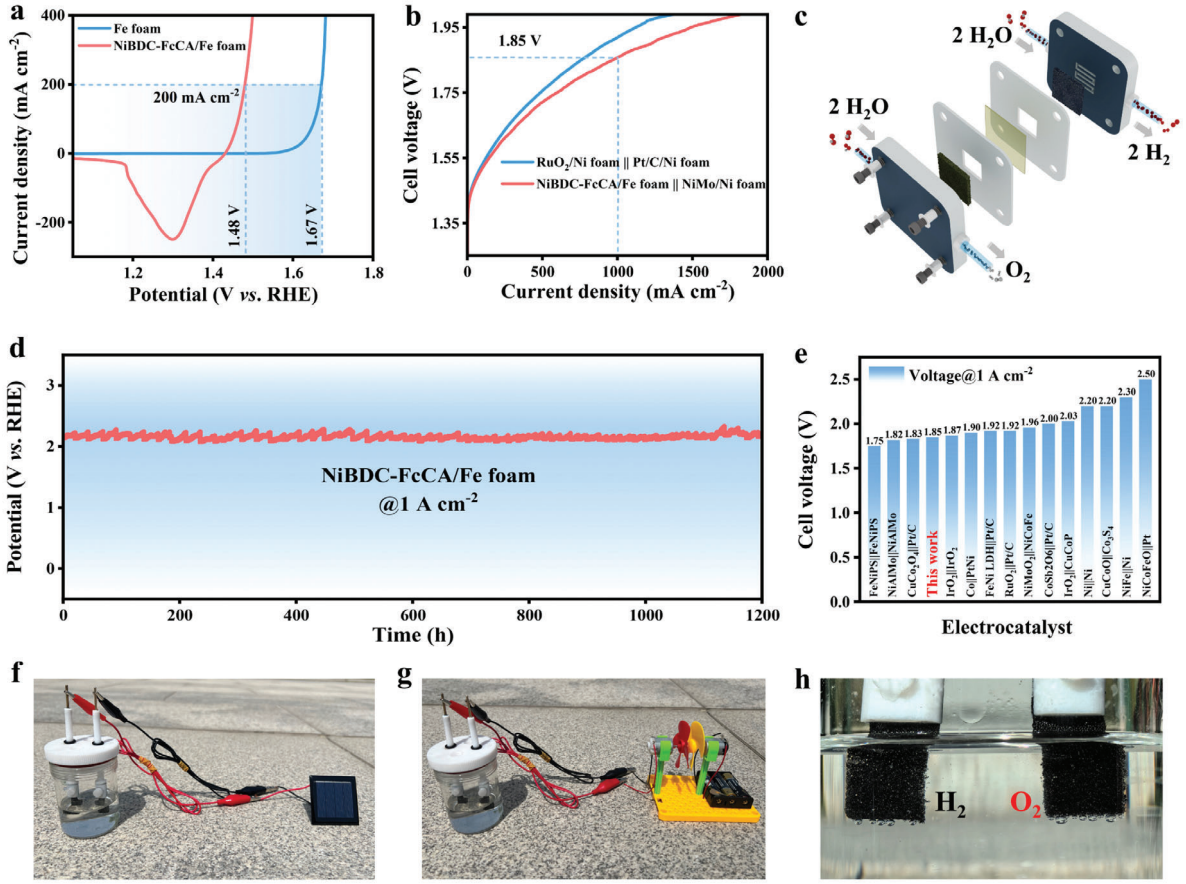


Figure 5. a) LSV curves obtained with NiBDC-FcCA/Fe foam and Fe foam. b) LSV curves obtained with NiBDC-FcCA/Fe foam||NiMo/Ni foam and commercial catalyst RuO_2/Ni foam|| Pt/C/Ni foam without *iR* compensation. c) Schematic diagram of electrolytic water reaction cell. d) Stability test of NiBDC-FcCA/Fe foam at a current density of 1 A cm^{-2} without *iR* compensation. e) Comparison of voltage at 1 A cm^{-2} of NiBDC-FcCA/Fe foam||NiMo/Ni foam and reported AEMWE cells. Simulation of f) solar-driven and g) wind-driven water splitting. h) O_2 and H_2 produced on the electrode surface.

(Supporting Information), the AEMWE cell containing NiBDC-FcCA/Fe foam||NiMo/Ni foam exhibits excellent durability, maintaining stable operation for at least 100 h at 1 A cm^{-2} (80 °C, 30 wt% KOH). It has been demonstrated leveraging renewable solar and wind energy for electrode reactions is crucial for achieving cost-effective water electrolysis.[18] Consequently, the solar and wind energy electrolysis water systems based on NiBDC-FcCA/Fe foam and NiMo/Ni foam electrodes were assembled (Figure 5f–g; Figure S35, Supporting Information). Notably, large amounts of O_2 and H_2 bubbles were observed on the electrode surfaces, signifying the significant application potential of the prepared electrode materials in water electrolysis (Figure 5h).

3. Conclusion

In summary, NiBDC-FcCA with defect structure was synthesized by introducing asymmetrically defective ligand FcCA into NiBDC. Designed structural features of pre-catalysts, oxygen vacancies and Fe doping, were preserved in multistep reconstruction, endowing reconstructed products with well-tuned electrical transport and efficient OER activity. The optimized NiBDC-FcCA possesses a low overpotential of 250 mV (200 mA cm^{-2}) and negligible activity change after 1200 h at a high current density of 1 A cm^{-2}. In addition, NiBDC-FcCA/Fe foam||NiMo/Ni foam exhibits even better performance than commercial catalysts in AEMWE. Further DFT calculations show that Fe doping reduces the adsorption strength of the oxygen intermediate with Ni sites, leading to a decrease in the ΔG of the RDS ($O^* \rightarrow OOH^*$), thereby enhancing the catalytic activity. This work paves the way for enhanced activity and stability of MOF-based OER catalysts for practical activity under simulated industrial conditions, driving the MOF-based electrochemical water splitting closer to broader implementation in real-world applications.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

metal-organic frameworks, oxygen evolution reaction, oxygen vacancy, structural reconstruction

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